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ICS2302 – Software engineering I Assignment

AUTOMATED SOFTWARE TESTING

Automated software testing involves using dedicated tools and scripts to run tests and simulate user actions on software programs to catch errors and bugs in the software development lifecycle. This involves rigorous repetition of unit testing (validating individual components/functions), regression testing (validating new features), validating API calls and load testing (validating performance over heavy traffic) before publishing.

They can be incorporated into Continuous Integration and Continuous Delivery/Deployment (CI/CD) pipeline i.e. Development Operations (DevOps).

Tools:

1. Selenium

Primarily used for web application testing.

Supports multiple programming languages (Java, Python, C#, JavaScript) and web browsers.

Provides cross-browser compatibility, making it a standard tool for UI automation.

Big community of software developers/testers.

2. Appium

Focused on mobile application testing for iOS and Android.

Allows testing of native, hybrid, and mobile web apps.

Uses the WebDriver protocol, enabling cross-platform test reuse.

3. Postman

Used for testing and developing APIs by providing intuitive interface for sending HTTP requests like POST and GET. (Backend testing)

Validation of endpoint API calls.

Usually to test login page.

4. Apache JMeter

Specializes in performance and load testing.

Simulates heavy traffic to evaluate server and application scalability.

Supports multiple protocols, including HTTP, FTP, and REST APIs.

5. Test Project

A free, community-driven platform for web, mobile, and API testing.

Provides a cloud-based environment for collaboration.

Features a script less recorder, making it accessible to non-programmers.

Automated software testing is essential in modern software development because it improves both efficiency and reliability. By using tools such as Selenium, Appium, Katalon Platform, Apache JMeter, Test Project, Postman, and Playwright, developers can identify bugs early, test different parts of the system, and ensure that applications perform well under various conditions.

References

- Ammann, P., & Offutt, J. (2016). *Introduction to Software Testing*. Cambridge University Press.
- Official websites : [Postman](#), [Katalon Platform](#), [Playwright](#) and [Appium](#)